

SAMADHAN MANE

AI/ML Engineering Intern | GenAI, RAG & LLM Systems | Python

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SUMMARY

Final-year Computer Engineering student building production-style GenAI and LLM systems. Winner of Neobim Hackathon 2026 with an agentic RAG platform, and open-source contributor to PipesHub AI. Strongest in retrieval pipelines, LLM tool calling and FastAPI backends, with hands-on PyTorch model training and measurable retrieval, latency and token-cost improvements.

TECHNICAL SKILLS

Languages: Python, SQL

Agentic AI & GenAI: Agentic AI, LangGraph, LangChain, LLM Tool Calling, RAG, Prompt Engineering, Reranking

ML, DL & NLP: PyTorch, TensorFlow, Keras, Scikit-learn, Machine Learning, Deep Learning, NLP, Hugging Face, Fine-tuning, Embeddings, NumPy, Pandas

Backend: FastAPI, REST APIs, JWT, RBAC

Databases: Pinecone, ChromaDB, MongoDB, Redis

Cloud & DevOps: AWS, Docker, Git, GitHub

PROJECTS

ABYSS – AI Due Diligence Platform (Agentic RAG)

Mar 2026 – Apr 2026

Python, FastAPI, React, TypeScript, Pinecone, Redis, PostgreSQL, Gemini 3.5 Flash Lite, Docker | [GitHub](#) | [Live Demo](#)

- Co-built ABYSS, a multi-tenant agentic RAG platform for business-document due diligence (query routing, hybrid retrieval, claim verification, citation-grounded answers); won 1st place at Neobim Hackathon 2026.
- Owned ingestion and retrieval: async OCR pipeline for 15+ file types, semantic chunking, MiniLM embeddings in Pinecone, and hybrid search with reranking that reached 0.88 MRR.
- Added Redis response caching so repeat queries return in 12 ms instead of ~3.7 s uncached.

BookMyHall – Campus Platform with Agentic AI Assistant

2025

LangGraph, LangChain, Gemini API, React, Node.js, Express.js, MongoDB, JWT, Tailwind CSS | [GitHub](#) | [Live Demo](#)

- Built a LangGraph agent with Gemini tool calling that turns natural-language requests into hall and vehicle bookings, canteen orders and maintenance tickets.
- Prevented double-booking with 5-minute slot locks and race-condition handling, so simultaneous requests for the same slot resolve safely.
- Shipped the full-stack platform (React, Express, MongoDB) with 10-tier RBAC, per-organization data isolation, multi-level approval chains and email notifications.

MOM-ai – Meeting Intelligence System (ASR + NLP + RAG)

2026

Python, PyTorch, Whisper, FLAN-T5, Mistral, LangChain, ChromaDB, Streamlit | [GitHub](#) | [Live Demo](#)

- Built a transcription and extraction pipeline (Whisper, FLAN-T5) that converts meetings into structured summaries, decisions and action items (0.864 BERTScore).
- Routed audio between Whisper and Sarvam AI for Hinglish speech, with LangChain + ChromaDB RAG that cut context token usage by ~65%.
- Trained Conv2D Autoencoder and VAE models in PyTorch on log-Mel spectrograms for acoustic representation learning (0.974 SSIM, 14 ms inference).

OPEN-SOURCE CONTRIBUTION

PipesHub AI – Reasoning-Effort Defaults for LLM Models

2026

Python, LLM Gateway, Unit Testing | [PR #3290](#) | [Merged PR #3828](#)

- Diagnosed reasoning models failing when no effort level was passed because the platform hardcoded a "high" fallback; authored PR #3290 adding request, model and platform-level defaults with unit tests.
- Maintainers later incorporated this approach into merged PR #3828 (persistence, validation, Bedrock, health checks), crediting #3290.

EDUCATION

B.Tech, Computer Engineering – MIT Academy of Engineering, Pune

2023 – 2027 (Exp.)

CGPA: 7.49/10

ACHIEVEMENTS

- Winner (1st Place), Neobim Hackathon 2026 – INR 1.8 lakh prize pool; 4-member team; for ABYSS.